

Integration - Set 1 - Solutions

Specification MM7.1-MM7.6.

Answer key

Question	Answer	Topic
1	F	Integration
2	C	Integration
3	C	Integration
4	B	Integration
5	D	Integration
6	E	Integration
7	F	Integration
8	A	Integration
9	C	Integration
10	E	Integration
11	D	Integration
12	C	Integration

Worked solutions

Question 1 INT-02 · Integration · spec MM7.1

Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

Answer: F

Roots 0 and 4; the curve is below the axis between them. $\int_0^4 (x^2 - 4x) dx = \frac{64}{3} - 32 = -\frac{32}{3}$, so the area is $\frac{32}{3}$.

Full method - how and why

Area between a curve and the axis. The integral is negative here because the region is below the axis - the area is its magnitude.

1 Find the limits: $x^2 - 4x = x(x - 4) = 0$ at $x = 0$ and $x = 4$.

why: The enclosed region runs between consecutive roots.

2 Note the curve is **below** the axis there: at $x = 2$, $y = -4$.

why: Checking one interior value tells you the sign of the integral in advance.

3 Integrate: $\int_0^4 (x^2 - 4x) dx = \left[\frac{x^3}{3} - 2x^2 \right]_0^4 = \frac{64}{3} - 32 = -\frac{32}{3}$.

why: $32 = \frac{96}{3}$, so $\frac{64 - 96}{3} = -\frac{32}{3}$.

4 Take the magnitude: the area is $\frac{32}{3}$.

why: Areas are positive; the negative sign only records that the region lies below the axis.

Where it goes wrong

- Giving $-\frac{32}{3}$ as the area - it is offered precisely to catch this.
- Using limits 0 and 2 (the vertex) instead of the two roots.

Takeaway Integrate between the roots, then take the absolute value: a definite integral is a signed area.

Question 2 INT-03 · Integration · spec MM7.3

Given that $\int_1^5 f(x) dx = 7$ and $\int_3^5 f(x) dx = 2$, find $\int_1^3 (2f(x) - 1) dx$.

Answer: C

$\int_1^3 f = 7 - 2 = 5$. So $2 \times 5 - (3 - 1) = 8$.

Full method - how and why

Combining integrals over adjacent intervals, plus the linearity rules. No function is ever needed.

1 Use $\int_1^5 = \int_1^3 + \int_3^5$: so $\int_1^3 f = 7 - 2 = 5$.

why: Integrals over adjacent ranges add; this is the only way to get at the interval you need.

2 Split the required integral: $\int_1^3 (2f(x) - 1) dx = 2\int_1^3 f(x) dx - \int_1^3 1 dx$.

why: Constants come out; sums split - both are properties of the integral.

3 Evaluate the second piece: $\int_1^3 1 dx = 3 - 1 = 2$.

why: Integrating 1 over an interval gives its length.

4 Combine: $2(5) - 2 = 8$.

why: Careful with the order of operations.

Where it goes wrong

- Forgetting the -1 term and answering 10.
- Using $\int_1^3 f = 7 + 2 = 9$ instead of subtracting.

Takeaway $\int_a^c = \int_a^b + \int_b^c$, constants factor out, and $\int_a^b 1 dx = b - a$.

Question 3 INT-04 · Integration · spec MM7.6

A curve has $\frac{dy}{dx} = 6x^2 - 4x + 1$ and passes through (1, 3). Find its equation.

Answer: C

$$y = 2x^3 - 2x^2 + x + c; \text{ at } (1, 3): 2 - 2 + 1 + c = 3 \Rightarrow c = 2.$$

Full method - how and why

Reverse differentiation plus a boundary condition to pin down the constant.

- 1 Integrate each term: $\int(6x^2 - 4x + 1)dx = 2x^3 - 2x^2 + x + c$.

why: Add one to each index and divide: $\frac{6x^3}{3} = 2x^3$, $\frac{-4x^2}{2} = -2x^2$.

- 2 Do not forget the constant c - it is exactly what the given point determines.

why: Every antiderivative differs by a constant, so one point selects the right curve.

- 3 Substitute (1, 3): $2 - 2 + 1 + c = 3$, so $c = 2$.

why: Evaluate the polynomial at $x = 1$ and set it equal to 3.

- 4 The curve is $y = 2x^3 - 2x^2 + x + 2$.

why: Check: differentiating gives back $6x^2 - 4x + 1$ ✓.

Where it goes wrong

- Omitting c altogether and choosing $y = 2x^3 - 2x^2 + x$.
- Arithmetic slip giving $c = 3$ - an offered option.

Takeaway $\frac{dy}{dx} = f(x)$ plus one point: integrate, then substitute to find c .

Question 4 INT-06 · Integration · spec MM7.5

The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?

Answer: B

$h = 1$: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$. The true value is $\frac{8}{3}$; the curve bends upwards (convex), so the chords lie above it and the rule overestimates.

Full method - how and why

Apply the trapezium rule and then decide over- or under-estimate from the curve's shape.

- 1 Two strips on $[0, 2]$ means $h = 1$ and ordinates at $x = 0, 1, 2$: $y = 0, 1, 4$.

why: $h = \frac{b-a}{n}$ with n the number of strips (not the number of ordinates).

- 2 Apply the rule: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$.

why: Only the interior ordinates are doubled.

- 3 Compare with the exact value $\int_0^2 x^2 dx = \frac{8}{3} \approx 2.67$.

why: The estimate 3 is larger.

- 4 Explain it: $y = x^2$ bends upwards (convex), so each chord lies **above** the curve and the trapezia overestimate.

why: Convex $\Rightarrow$ overestimate; concave $\Rightarrow$ underestimate. This is the reasoning the question wants.

Where it goes wrong

- Forgetting to double the interior ordinate, giving $\frac{5}{2}$ - offered twice as a distractor.
- Saying 'underestimate' because the trapezia look like they cut corners - for a convex curve they do the opposite.

Takeaway Trapezium rule: $\frac{h}{2}[y_0 + 2(\text{middles}) + y_n]$; convex curves are overestimated.

Question 5 INT-07 · Integration · spec MM7.3

Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

Answer: D

$$a^2 + a = 12 \Rightarrow (a + 4)(a - 3) = 0 \Rightarrow a = 3.$$

Full method - how and why

A definite integral with an unknown limit. Integrate, substitute, and solve the resulting equation.

- 1 Integrate: $\int_0^a (2x + 1) dx = [x^2 + x]_0^a = a^2 + a$.
why: The lower limit 0 contributes nothing, which keeps this clean.
- 2 Set equal to 12: $a^2 + a - 12 = 0$.
why: Bring everything to one side.
- 3 Factorise: $(a + 4)(a - 3) = 0$, so $a = -4$ or $a = 3$.
why: Two numbers multiplying to -12 and adding to 1.
- 4 Apply the condition $a > 0$: $a = 3$.
why: The question's restriction exists to select one root.

Where it goes wrong

- Forgetting the '+x' from integrating the constant 1, giving $a = \sqrt{12}$ - an offered option.
- Giving both roots when only the positive one is allowed.

Takeaway An unknown limit becomes an equation after integration; use any stated sign condition to choose the root.

Question 6 INT-08 · Integration · spec MM7.4

Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.

Answer: E

By the Fundamental Theorem of Calculus $F'(x) = x^2 - 3x$, so $F'(3) = 9 - 9 = 0$.

Full method - how and why

A direct test of the Fundamental Theorem of Calculus: differentiating an integral with a variable upper limit gives back the integrand.

- 1 Apply the FTC: $\frac{d}{dx} \int_2^x g(t) dt = g(x)$, so $F'(x) = x^2 - 3x$.
why: The lower limit is a constant, so it contributes nothing to the derivative.
- 2 Substitute $x = 3$: $F'(3) = 9 - 9 = 0$.
why: Just evaluate the integrand at 3 - no integration is needed at all.

Where it goes wrong

- Actually integrating and then differentiating - correct but slow, and error-prone.
- Substituting $t = 3$ into an antiderivative and answering $-\frac{7}{6}$ or $\frac{9}{2}$, both offered.

Takeaway $\frac{d}{dx} \int_a^x f(t) dt = f(x)$: differentiation undoes integration, so just evaluate the integrand.

Question 7 INT-09 · Integration · spec MM7.1

Which of the following statements is/are true?

I $\int_{-2}^2 x^3 dx = 0$

II The area between $y = x^3$ and the x -axis from $x = -2$ to $x = 2$ is 0.

III $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

Answer: F

I: odd function over a symmetric interval, so 0. II is false: the area is $2 \int_0^2 x^3 dx = 8$; the integral is 0 because the parts cancel. III: even function, true.

Full method - how and why

Three statements distinguishing **signed integrals** from **areas**, using odd and even symmetry.

1 I: x^3 is odd, so the contributions from $[-2, 0]$ and $[0, 2]$ cancel exactly: the integral is 0. **True.**

why: For odd f , $\int_{-a}^a f = 0$ always.

2 II: the **area** counts both pieces positively: $2 \int_0^2 x^3 dx = 2 \times 4 = 8 \neq 0$. **False.**

why: Area and integral differ precisely when the curve crosses the axis - this is the distinction the question is testing.

3 III: x^2 is even, so the two halves are equal: $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$. **True.**

why: For even f , $\int_{-a}^a f = 2 \int_0^a f$.

4 I and III only.

why: Match to the option list.

Where it goes wrong

- Treating 'integral' and 'area' as synonyms and marking II true.
- Thinking odd symmetry also halves the area.

Takeaway Odd function over a symmetric interval: integral 0 but area positive. Even function: integral is twice the half.

Question 8 INT-10 · Integration · spec MM7.2

Find $\int \left(x + \frac{1}{x}\right)^2 dx$.

Answer: A

$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + x^{-2}$, which integrates to $\frac{x^3}{3} + 2x - x^{-1} + c$. $\left(\frac{1}{3}\left(x + \frac{1}{x}\right)^3 + c\right)$ is a common error: you cannot integrate a power of a bracket that way.)

Full method - how and why

Expand before integrating - there is no rule for integrating a power of a bracket on this specification.

1 Expand: $\left(x + \frac{1}{x}\right)^2 = x^2 + 2\left(x \cdot \frac{1}{x}\right) + \frac{1}{x^2} = x^2 + 2 + x^{-2}$.

why: The cross term is $2 \times x \times \frac{1}{x} = 2$, a constant - easy to lose.

2 Integrate term by term: $\frac{x^3}{3} + 2x + \frac{x^{-1}}{-1} = \frac{x^3}{3} + 2x - \frac{1}{x}$.

why: $\int x^{-2} dx = \frac{x^{-1}}{-1} = -\frac{1}{x}$; the index -2 is allowed because it is not -1 .

3 Add the constant: $\frac{x^3}{3} + 2x - \frac{1}{x} + c$.

why: Indefinite integrals always need $+c$.

Where it goes wrong

- Choosing $\frac{1}{3}\left(x + \frac{1}{x}\right)^3 + c$ - you cannot integrate a bracket by raising its power.
- Sign error giving $+\frac{1}{x}$, or dropping the $2x$ from the cross term.

Takeaway Expand brackets fully first; $\int x^{-2} dx = -x^{-1}$, and the cross term of a square is easy to forget.

Question 9 INT-12 · Integration · spec MM7.2

Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

Answer: C

$$[3x^{1/3}]_1^8 = 3(2 - 1) = 3.$$

Full method - how and why

A negative fractional index. Add one to it, and choose limits that make the cube root easy.

1 Add one to the index: $-\frac{2}{3} + 1 = \frac{1}{3}$, and divide by $\frac{1}{3}$ (i.e. multiply by 3): $\int x^{-2/3} dx = 3x^{1/3}$.
why: Dividing by a fraction multiplies by its reciprocal.

2 Evaluate at the limits: $3(8^{1/3}) - 3(1^{1/3}) = 3(2) - 3(1)$.
why: $8^{1/3} = 2$ - the limits were chosen to be perfect cubes.

3 Subtract: $6 - 3 = 3$.
why: Simple arithmetic.

Where it goes wrong

- Forgetting the factor 3 and answering 1 or 2.
- Adding one to the index the wrong way and using $x^{-5/3}$.

Takeaway $\int x^n dx = \frac{x^{n+1}}{n+1}$ works for negative and fractional n (except $n = -1$); dividing by $\frac{1}{3}$ multiplies by 3.

Question 10 INT-13 · Integration · spec MM7.3

$f(x) = ax + b$ satisfies $\int_0^1 f(x) dx = 2$ and $\int_0^2 f(x) dx = 6$. Find $f(1)$.

Answer: E

$\frac{a}{2} + b = 2$ and $2a + 2b = 6$, so $a + b = 3$; subtracting the first equation from this gives $\frac{a}{2} = 1$, so $a = 2$, $b = 1$ and $f(1) = 3$.

Full method - how and why

Two integral conditions on a linear function give two equations in a and b .

1 Integrate in general: $\int_0^k (ax + b) dx = \frac{ak^2}{2} + bk$.
why: Doing it once with a general limit saves repeating the work.

2 $k = 1$: $\frac{a}{2} + b = 2$. $k = 2$: $2a + 2b = 6$, i.e. $a + b = 3$.
why: Substitute each limit and simplify the second by dividing by 2.

3 Subtract the first from the second: $\frac{a}{2} = 1$, so $a = 2$, and then $b = 1$.
why: Elimination: $a - \frac{a}{2} = \frac{a}{2}$ and $3 - 2 = 1$.

4 Compute $f(1) = a + b = 3$.
why: Conveniently, $f(1) = a + b$, which the second equation already gave.

Where it goes wrong

- Forgetting the $\frac{1}{2}$ in $\frac{ak^2}{2}$ and getting a different pair.
- Answering 2 (the value of the first integral) instead of $f(1)$.

Takeaway Integrate once with a general limit, then substitute each condition to get simultaneous equations.

Question 11 INT-16 · Integration · spec MM7.4

Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

Answer: D

$$10 + 24 - (-10) = 44.$$

Full method - how and why

Three integral rules in one: linearity, the integral of $3x$, and the effect of swapping limits.

1 Split the first integral: $\int_0^4 (f(x) + 3x) dx = \int_0^4 f(x) dx + \int_0^4 3x dx = 10 + \int_0^4 3x dx$.

why: Sums split into separate integrals.

2 Compute $\int_0^4 3x dx = \left[\frac{3x^2}{2} \right]_0^4 = 24$.

why: $\frac{3 \times 16}{2} = 24$.

3 Handle the second term: $\int_4^0 f(x) dx = -\int_0^4 f(x) dx = -10$.

why: Reversing the limits reverses the sign - the key fact here.

4 Combine: $(10 + 24) - (-10) = 44$.

why: Subtracting a negative adds.

Where it goes wrong

- Treating $\int_4^0 f$ as $+10$ and answering 24.
- Forgetting the $3x$ term, giving 20.

Takeaway $\int_b^a = -\int_a^b$; watch for reversed limits used as a deliberate trap.

Question 12 INT-17 · Integration · spec MM7.6

The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y-intercept of the curve.

Answer: C

$$y = 2x^{3/2} + c; 20 = 16 + c \Rightarrow c = 4. \text{ The y-intercept is 4.}$$

Full method - how and why

Integrate the gradient function, use the given point for c , then read the intercept.

1 Integrate: $\int 3x^{1/2} dx = 3 \times \frac{x^{3/2}}{3/2} = 2x^{3/2}$, so $y = 2x^{3/2} + c$.

why: $3 \div \frac{3}{2} = 2$.

2 Substitute $(4, 20)$: $4^{3/2} = 8$, so $2(8) + c = 20$, giving $c = 4$.

why: $4^{3/2} = (\sqrt{4})^3 = 8$.

3 The y-intercept is the value at $x = 0$: $y = 0 + 4 = 4$.

why: For this curve the intercept is just c .

Where it goes wrong

- Using $4^{3/2} = 6$ and getting $c = 8$ - an offered option.
- Reporting 20 (the given y-value) as the intercept.

Takeaway The constant of integration **is** the y-intercept when the rest of the expression vanishes at $x = 0$.